

LAYER 1:Environment (Physical environment & Signal Channel)



Generation of physical ground truth and channel parameters. Modular scene and model selection

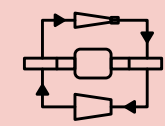
Ray-tracing

Empirical
(e.g. TDL-A)

Reflection-
only

Channel Truth
(delay,amplitude)

LAYER 2:Hardware Impairments (Physical limitations)

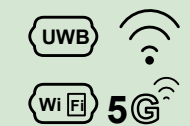


Injecting realistic hardware defects and limitations. Configurable pipeline for RF non-idealities.

Hardware Impairment Injection
Pipeline
(CFO, Filter, SFP, I/Q imbalance, etc.)

Channel Truth
(impaired)

LAYER 3:Transceiver (Multi-protocol Signal)



Protocol-specific baseband processing. Fair comparison across different standards.

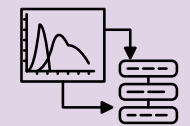
UWB Stack
(e.g. IEEE 802.15.4z)

WiFi FTM Stack
(e.g. IEEE 802.11az)

5G PRS Stack
(e.g. Release-16/17)

Radio Observation
(CIR)

LAYER 4:Algorithms (Ranging & Multipath Detection)



Structured competitive environment for various detection algorithms. Testing new designs.

First-Path
Detection
(e.g. Threshold)

Multipath Detection
(e.g. SAGE, MUSIC)

Range Estimates &
Multipath Results

LAYER 5:Evaluation (Monte Carlo Statistics & Metrics)



Comparing results to ground truth. Generating performance metrics.

Error
Calculation

Monte Carlo
Simulator

Performance
Charts

Evaluation Results &
Performance Graphics